

mlxprivhost - NIC Configuration by the Host Restriction Tool (Zero Trust Mode)

mlxprivhost enables the user to restrict the hosts from managing the device in case the BlueField DPU is in a Zero Trust environment, and the host cannot be considered “trusted”.

mlxprivhost is supported in Linux only.

mlxprivhost is not supported in ESXi 7.0.

This utility is supported in BlueField devices only.

mlxprivhost Synopsis

```
1  mlxprivhost [OPTIONS] <command> [parameters...]
```

- Restrict configuration takes effect immediately, but **disabling/enabling RShim** requires power cycle
- A zero-trust (restricted) host will not be able to perform operations that can compromise the DPU, such as:
- For Multi-host systems, the tool is compatible with firmware versions starting from xx.31.10xx and later

where:



-h, --help	Shows this help message and exit
-v, --version	Shows program's version number and exit
-d <dev>, --device <dev>	Device to work with.

--disable_rshim	When TRUE, the host does not have an RSHIM function to access the embedded CPU registers (power cycle is required to apply changes)
--disable_tracer	When TRUE, the host will not be allowed to own the Tracer (requires FW reset to be applied)
--disable_counter_rd	When TRUE, the host will not be allowed to read Physical port counters (requires FW reset to be applied)
--disable_port_owner	When TRUE, the host will not be allowed to be Port Owner (requires FW reset to be applied)
r,restrict	Set all external hosts as zero-trust (restricted) except of the one that called the command
p,privilege	Set all external hosts privileged except the one that called the command
q,query	From external HOST: query the status of the host From Embedded ARM CPU: query the status of all external hosts.
-f, --full	Run with query command for high verbosity level - valid from embedded ARM CPU only.

Example of mlxprivhost:

- Enabling Zero-Trust host (Full Host Restriction - Embedded ARM CPU Only):

```
1 mlxprivhost -d /dev/mst/mt41682_pciconf0 r --disable_rshim --disable_tracer
```

```
1 mlxprivhost -d /dev/mst/mt41682_pciconf0 p
```

- Query the status of the host\hosts (the full flag valid for embedded ARM CPU Only):

```
mlxprivhost -d /dev/mst/mt41682_pciconf0 q --full
Host configurations
-----
host index      : 0          1          2          3
level           : PRIVILEGED PRIVILEGED PRIVILEGED PRIVILEGED

Port functions status:
-----
disable_rshim   : FALSE     FALSE     FALSE     FALSE
10
```

10					
11	disable_tracer	: FALSE	FALSE	FALSE	FALSE
12	disable_port_owner	: FALSE	FALSE	FALSE	FALSE
	disable_counter_rd	: FALSE	FALSE	FALSE	FALSE

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